

Does Machine Learning reconstruct missing sunspots and forecast a new solar minimum?

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Received 29 October 2020; received in revised form 1 March 2021; accepted 19 March 2021

Available online 30 March 2021

Abstract

The retrodiction and prediction of solar activity are two closely-related problems in dynamo theory. We applied Machine Learning (ML) algorithms and analyses to the World Data Center's newly constructed annual sunspot time series (1700–2019; Version 2.0). This provides a unique model that gives insights into the various patterns of the Sun's magnetic dynamo that drives solar activity maxima and minima. We found that the variability in the ~ 11 -year Sunspot Cycle is closely connected with 120-year oscillatory magnetic activity variations. We also identified a previously under-reported 5.5 year periodicity in the sunspot record. This 5.5-year pattern is modulated by the 120-year oscillation and appears to influence the shape and energy/power content of individual 11-year cycles. Our ML algorithm was trained to recognize such underlying patterns and provides a convincing hindcast of the full sunspot record from 1700 to 2019. It also suggests the possibility of missing sunspots during Sunspot Cycles -1 , 0 , and 1 (ca. 1730s–1760s). In addition, our ML model forecasts a new phase of extended solar minima that began prior to Sunspot Cycle 24 (ca. 2008–2019) and will persist until Sunspot Cycle 27 (ca. 2050 or so). Our ML Bayesian model forecasts a peak annual sunspot number (SSN) of 95 with a probable range of 80–115 for Cycle 25 between 2023 and 2025.

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Keywords: Forecasting sunspot cycles; Missing sunspots; Future sunspot activity minima; Machine Learning

1. Introduction

Variations in solar activity have fundamental impacts for Earth's climate and for all life. Understanding how sunspot activity and, hence, how the general solar magnetic activity varies in time has been the focus of much research. This deficit in knowledge persists despite the clear demonstration of the 11-yr-like periodic oscillations over the available 400-yr sunspot history. The need to foresee the future of solar activity and to account for its wide-ranging impacts could not be more urgent. For example, the useful life of a micro-satellite or Cube-Sat will depend

on the future magnitude of solar activity. If this technology is to be properly managed, long-term reliable forecasts of solar activity will be an important prerequisite.

Over the past 60–70 years, the challenge to forecast sunspot activity has focused on methods and algorithms ranging from internal and external precursors, dynamo models, and nonlinear attractor analyses to neural networks (see McNish and Lincoln, 1949; Petrovay, 2020). The consensus-building efforts of the international team forecasting solar Cycle 25 at the NOAA-NASA panel (i.e., released around December 9, 2019)¹ had proposed a peak

¹ <https://www.swpc.noaa.gov/news/solar-cycle-25-forecast-update>

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Table 1

Statistical characteristics for the International Sunspot Number Version 2.0 record from the World Data Center.

	Statistical Characteristics	
	mean value <SSN>	standard deviation σ
Annual Sunspot Number (1700–2019)	78.7	62.1
Bayesian ML Model of SSN (1700–2019) ^a	80.7	63.8
Long-term Annual Sunspot Number (1700–2100) ^a	83.2	64.2

^aResults obtained in this paper.

sunspot number of 115 to occur in July of 2025; a number similar to the peak of Cycle 24. Regardless of the ultimate value, no method has consistently offered the best forecast of non-linear dynamics of future solar activity cycles.

The long-range forecasting method, allowing up to 15 future solar cycles (i.e., 2008–2176), adopted by Hiremath (2008) should be highlighted for the fact that the author's early prediction of a nearly equal maximum of 110 ± 11 sunspots for both solar Cycles 24 and 25 is proven to be relatively insightful. This early prediction for Cycle 25 by Hiremath agrees favorably with the later forecasts offered by Podladchikova et al. (2017), Bhowmik and Nandy (2018), and Singh and Bhargawa (2019) but disagrees substantially with the strong Cycle 25 with peak sunspot numbers of 180 ± 60 suggested by Pesnell (2018) or even the medium-range peak value of 131 to 134 ± 11 estimated by Biso et al. (2020).²

In this paper, we report a novel method based on the solar magnetic patterns deduced from a Bayesian Machine Learning model, not only for the upcoming solar Cycle 25, but also for several subsequent cycles. Our possibility of success exists because of the long time horizon of our proposed precursor is based upon the 120-yr timescale derived from our original reanalyses of the sunspot record.

2. Data and method

Here, we briefly describe the sunspot record and the unique methodologies applied to reproduce it.

2.1. Sunspot data

We used Version 2.0 of the International Sunspot Number (SSN) from the World Data Center Sunspot Index and Long-term Solar Observations (WDC-SILSO), Royal Observatory of Belgium, Brussels³. The key documentation for these reconstructions and formal statistical uncertainties are discussed in Clette et al. (2014) and Dudok et al. (2016), respectively.

This annual SSN time series covers the period from 1700 to 2019. We utilized statistical characteristics (mean value and standard deviation) of the annual SSN measurements

as a criterion to quantify the low or high solar activity of the next solar cycle (Table 1).

2.2. Wavelet spectral analysis

First, we find possible solar magnetic patterns that modulate the amplitude of sunspot numbers and that induce variations in solar activity. Although different methods exist to find correlated patterns in time series, we have chosen the Wavelet Transform (WT, see Torrence and Compo, 1998; Grinsted et al., 2004; Soon et al., 2019; Velasco Herrera et al., 2015; Soon et al., 2014, for more details about the method) in analysing the SSN time series record because the wavelet spectra allows identification of intrinsic patterns of the phenomenon and facilitates the discovery of the characteristics of the phenomena's source; for example, the inner workings of the Sun's magnetic dynamo (see e.g., Soon et al., 2019; Frick et al., 2020).

Wavelet analysis began concomitantly with the philosophy of quantum physics by the work of Alfred Haar in 1909. But wavelets went practically unnoticed until 1984 when Grossmann and Morlet (1984) wrote their seminal paper. We use the Morlet wavelet mother in the WT because of its high precision in resolving the patterns (periodicities) contained in the sunspot records and because it is a complex function that allows us to deduce the information on phase of the dominant timescale of the solar magnetic patterns.

The wavelet transform of a discrete time series y_n (e.g., the annual sunspot number) is defined by Torrence and Compo (1998) as:

$$\mathbf{W}_n(s) = \sum_{n'=0}^{N-1} y_n \psi_o * \left(\frac{(n'-n)}{s} \delta t \right) \quad (1)$$

where s is the scale, n is the translation parameter (sliding in time) and the $(*)$ denotes complex conjugation.

The decomposition of a signal (y_n) in channel or bandwidth can be obtained from inverse wavelet (Torrence and Compo, 1998) as:

$$y_n = \frac{\delta_j \delta t^{1/2}}{C_\delta \psi_o(0)} \sum_{j=j_1}^{j_2} \frac{Re(\mathbf{W}_n(s_j))}{s_j^{1/2}} \quad (2)$$

where j_1 and j_2 define the scale range of the specified spectral bands, $\psi_o(0)$ is an energy normalization factor, C_δ is a reconstruction factor, and δ_j is a factor for scale averaging.

² We note that the authors referred to their prediction for the peak of Cycle 25 as “mini solar maximum”.

³ <http://www.sidc.be/silso/datafiles>

For Morlet wavelet, $\delta_j = 0.6$, $C_\delta = 0.776$, and $\psi_o(0) = \pi^{-1/4}$.

The input data in the WT are the annual SSN time series from 1700 to 2019. The WT has 3 outputs (as shown in Figs. 1, 2 and 6); the global (or time-averaged) frequency spectrum which shows the periodicities (patterns) existing in SSN record left panel; the local spectrum that shows the evolution over time of these periodicities as well their phases center panel, and the amplitude and phase of the dominant pattern shown in the global wavelet bottom

panel. These time series are the input data in ML algorithms for hindcasting and forecasting sunspot cycles.

2.3. Machine learning algorithms for probabilistic hindcasting and forecasting of time series

In recent years, the use of Artificial Intelligence (AI) in forecasting astronomical and astrophysical phenomena has increased rapidly (see e.g., Fluke and Jacobs, 2019). Specifically, Deep Learning (DL) and Machine Learning

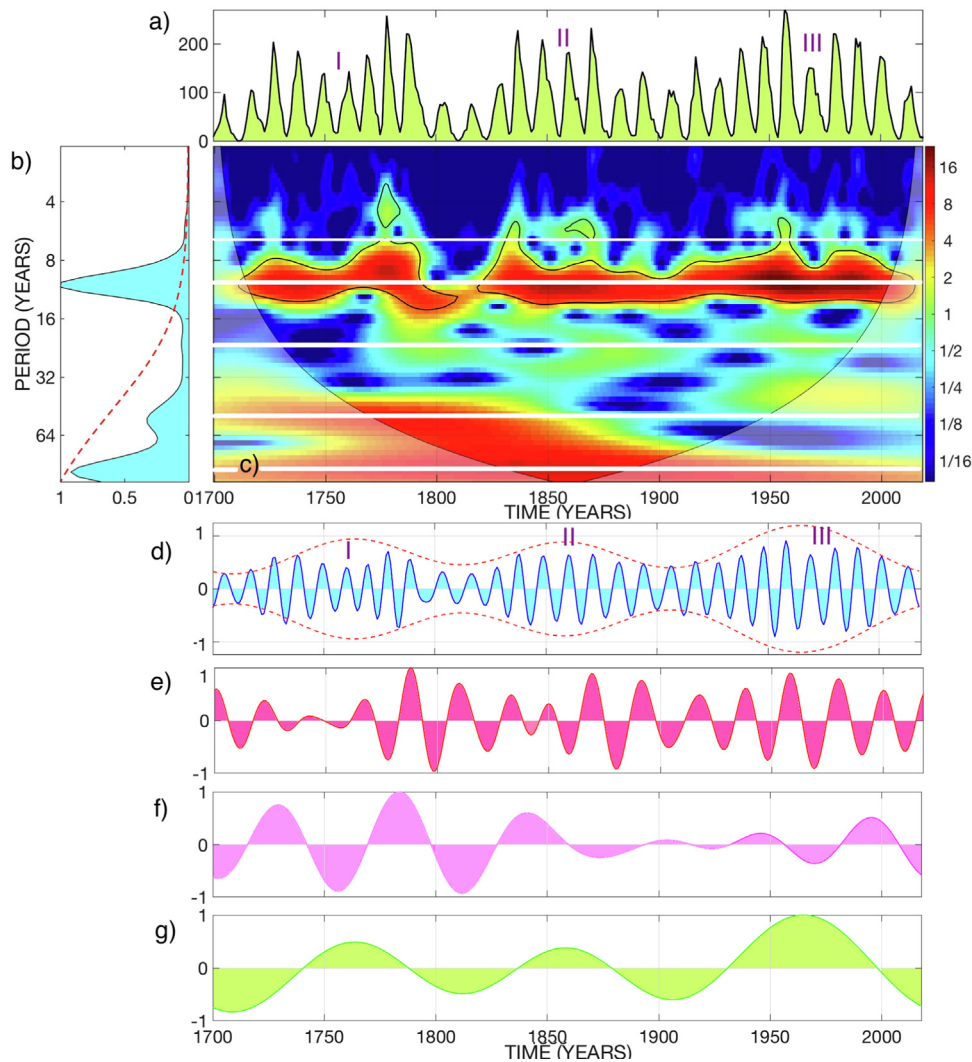


Fig. 1. Results of the time–frequency wavelet spectral analysis. (a) Annual sunspot time series records from 1700–2019. (b) The global time-averaged wavelet period (left-hand panel) with the red dashed line indicating the 95% confidence level drawn from a red noise spectrum (red dashed line). (c) The Morlet wavelet power spectral density (MWPSD) in arbitrary units adopting the red-green-blue colour scales. The cone of influence shows the possible edge effects in the MWPSD (i.e., the U-shaped curves (outside of which the spectral information can be considered unreliable). In addition, a horizontal white line across panel 1c marks the significant 5.5-yr period which is highlighted and discussed in Fig. 2 and Section 3.2). Four additional horizontal white lines are plotted to correspond to the time variation of the four periods plotted under Figs. 1d, 1e, 1f and 1g, respectively. (d) Periodic variations of the 11-year solar cycle (e.g., the Schwabe cycle) with the envelope of modulation by the 120-yr oscillations (pink curves). (e) Periodic variations of the solar cycle of 22 years (Hale cycle). (f) The periodic variations of the 60-year solar cycle (i.e., variously known as the Yoshimura-Gleissberg cycle; see discussion in text and Soon et al. 2014). (g) Periodic variations of the solar cycle of 120 years (i.e., variously known as the Yoshimura-Gleissberg cycle; see discussion in the text and Soon et al. 2014). Roman numerals ‘I’, ‘II’, and ‘III’ in panels (a) and (d) denote the Wolf’s cycle discussed in the main text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

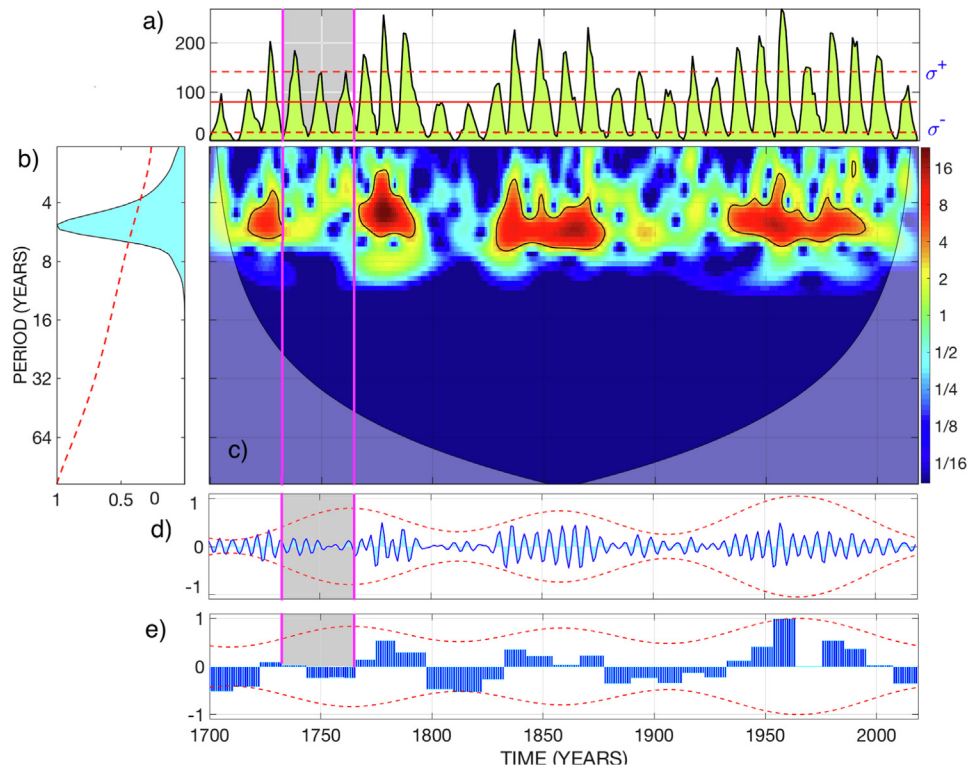


Fig. 2. Time–frequency wavelet spectrum of annual sunspot time series records from 1700–2019. Horizontal solid (middle) and two dashed (upper and lower) red lines are the mean and standard deviation of the sunspot number statistics reported in Table 1. Panels (a, b, and c) present similar information as described in Fig. 1. Panel (d) shows the periodic variations of the 5.5-yr solar cycle. Panel (e) shows the temporal power anomaly for each 11-yr solar cycle. The pink curves in both panels (d) and (e) are the envelope of modulation by the 120-yr oscillations. Vertical pink lines across panels (a), (c), (d) and (e) mark solar Cycles -1 , 0 and 1 (ca. 1733–1767) that have significant uncertainties in the reconstructed sunspot number estimates and sunspot numbers during this interval may have been underestimated (see Hoyt and Schatten, 1998a; Hoyt and Schatten, 1998b; Clette et al., 2014, and the text for more detailed discussion).

(ML) algorithms are very important for time series analysis. ML and DL have very sophisticated and complex algorithms for forecasting time series (e.g., Lewis, 2016; Nielsen, 2019).

Here, we explore the capacity of ML as a powerful tool to understand the complex nature of solar magnetic activity. Our primary goal is to improve our understanding of the magnetic patterns that induce high and low solar cycles and because this is the most essential prerequisite to allow the prediction of long-term variability of sunspots. Sunspot forecasts with published algorithms and methodologies have only been able to forecast one or two solar cycles ahead and the results have not been entirely satisfactory (see Petrovay, 2020, for a full review of the solar cycle forecasting algorithms and their performance, including initial forecasts for solar Cycle 25).

We have used a novel methodology with different combination of ML algorithms to find magnetic precursors to forecast both the amplitude and shape of the following solar cycles and to identify the magnetic patterns of high or low solar activity (both secular and 11-yr-like solar cycle). These patterns are essential to allow the forecasting of solar variability on longer time horizons. For time series analysis, ML has both supervised and unsupervised algorithms. Although different types of

models exist for hindcasting and forecasting time series, any model is only an approximation of reality and therefore its quality in estimating the parameter to forecast is limited by a principle of uncertainty (Velasco Herrera et al., 2015).

2.3.1. Bayesian inference for Least-Squares Support-Vector Machines (LS-SVM) regression

Regardless of the method used to offer a forecast sunspot activity cycles, at least two types of data should be considered:

1. The actual recorded data of the Version 2.0 SSN.
2. The “modelled” SSN time series. Such a model can be a filter that eliminates high frequencies (for example, the so-called Resistor-inductor(L)-Capacitor (RLC) or Resistor–Capacitor (RC), electric oscillator model), a bandpass model (for example, using Wavelet, Fourier, etc.), a ML or DL model, a probabilistic model, etc. The level of confidence will depend on the model that is used.

We use a Bayesian inference for the adopted LS-SVM regression model (see Suykens et al., 2005, for technical questions about the method) mainly because it is impossible to forecast the “exact” number of sunspots in the following solar cycles. The Bayesian inference model is based on Bayes’s theorem (Bayes, 1763.) stated as follows:

$$p(H|D) = \frac{p(D|H)}{p(D)} p(H) \quad (3)$$

where H is the LS-SVM regression model and D is training data (in our case is the Version 2.0 SSN).

In other words,

$$\text{Posterior} = \frac{\text{Likelihood}}{\text{Evidence}} \text{Prior}$$

or the likelihood at a certain level equals the evidence at the previous level (i.e., the parameters at different levels) and subsequent levels are linked to each other. Bayes's theorem is used to deduce the optimal parameters of the LS-SVM model which will be described later. The LS-SVM method, including the radial basis function (RBF) kernel, was used because its efficacy in forecasting Total Solar Irradiance at high spectral and temporal resolutions according to our past experience (Velasco Herrera et al., 2015). Ultimately, we use the Bayesian inference ML model obtained from the SSN time series from 1700 to 2019 to create probabilistic models of the solar cycles (see Fig. 3).

2.3.2. Non-linear AutoRegressive eXogenous (NARX) model

Specifically, we use the Non-linear AutoRegressive eXogenous (NARX) model to create models of solar cycles hindcasting and forecasting. The NARX model is an expansion of past input and output terms and the essence of the NARX model is that past outputs are included in the expansion and is defined as:

$$\hat{y}(k) = f[y(k-1), \dots, y(k-p), u(k-1), \dots, u(k-q)] \quad (4)$$

where $\hat{y}$ is the estimated SSN at time “ k ”, y and u denotes the output and input data, respectively. The order of the

system is determined by the input and output values p and q , representing the number of lags.

The function f in Eq. (4) is non-analytic and can be implemented using different approaches (see Ayala Solares et al., 2016, for more details about the method). We modeled f with the LS-SVM method (Suykens et al., 2005) as:

$$f = \sum_{k=1}^n \Omega^k u_k + b \quad (5)$$

where u_k denotes the input data at time k (discrete time index from $k = 1, \dots, n$), Ω^k is the weighting factor that can, in turn, have functional dependence on u_k , $\Omega^k = \Omega^k(u_k)$, and b is the “bias” term.

At each step “ k ”, the input data in Eq. (5) is the value of the Bayesian inference model of the SSN at time “ k ”. The output at time “ k ” is the estimated value of the SSN ($\hat{y}$).

2.3.3. Wavelet-LS-SVM algorithms for the estimation of solar cycles for a multi-channel system

Here we describe a novel methodology to estimate solar Cycles 25–32 (i.e., forecast) and solar Cycles from -1 to 1 (i.e., hindcast). We combine the algorithms of the wavelet transform and Machine Learning, we further follow these iterative steps:

- I Use wavelet transform (Eq. (1)) to find the periodicities of the Version 2.0 SSN record. The results are shown in Fig. 1b.
- II The decomposition of the Version 2.0 SSN record in time series called “channels” with the periodicities obtained in step I, can be obtained using inverse wavelet (Eq. (2)). The results are shown in Fig. 1(d)–(g).

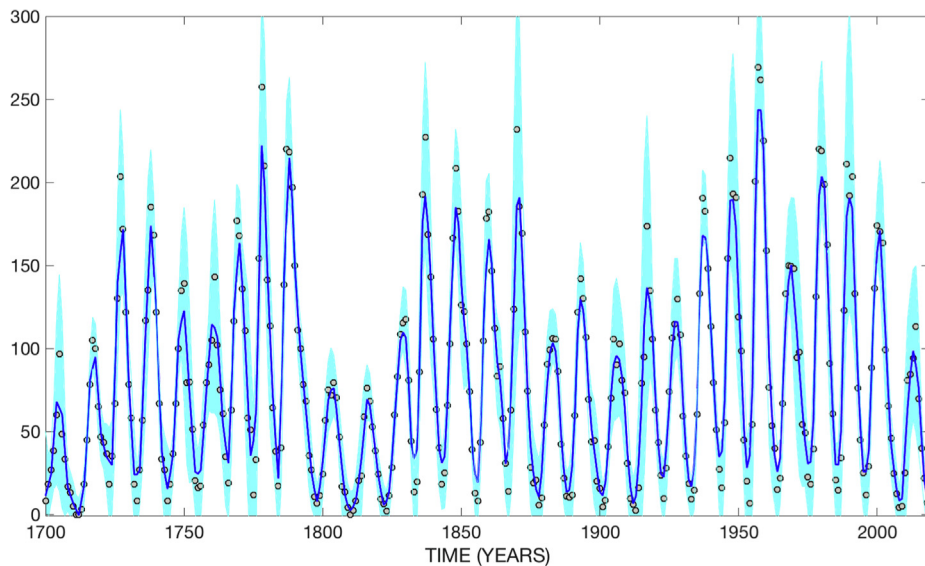


Fig. 3. Our Bayesian inference of LS-SVM model (blue line) compared with the actual observed annual sunspot time series (gray dots) from 1700–2019. The blue shaded area represents the 95% confidence intervals of the Bayesian model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- III Use Eq. (3) to obtain a Bayesian inference model from the Version 2.0 SSN time series between 1700 and 2019 (blue line in Fig. 3). This Bayesian inference model will be used in estimating SSN values of the solar Cycles 25–32.
- IV Find in the Bayesian inference model each of the periodicities obtained in the step (I) and decompose this Bayesian inference model in time series that we called “channels”, with the inverse wavelet (Eq. 2). Each of these “channel” will be the input in the NARX model (Eq. 4) to make the forecast of solar Cycles 25–32
- V Selection of the model lags p and q for each “channel”, i.e., time series with the periodicities obtained in the step (IV).
- VI Use the K -fold cross validation for the training, validation, testing and deduction of the parameters of the NARX model.
- VII Determination of the weight and bias for each “channel” analyzed.
- VIII Integrate all channels to obtain the Bayesian inference model that will estimate two consecutive solar cycles using Eq. (4).
- IX Test of the accuracy of the estimate of two solar cycles. As an example, in Section 4.2 the test of the accuracy of the estimate for solar Cycles 23 and 24 is described. The result is shown in Fig. 4.
- X Test of the cost function. We used the mean squared error (MSE). If this function was small enough, stop and go to the next step (XI). Otherwise, return to step (V) and change the parameters “ p ” and/or “ q ”.
- XI Use the wavelet transform to help determine if the periodicities of the estimated solar cycles have the same periodicities obtained in (III). If yes, then with these new data (i.e. with the input data and these

two new solar cycles), go to step (VIII) to calculate the next two solar cycles. Otherwise, repeat from step (V).

- XII After obtaining N forecasts and in order to eliminate extreme values, perform an arithmetic average. Then use the “average model” as input in Eq. (3) to obtain a Bayesian model of solar Cycles 25–32. The result is shown in Fig. 5.

For training, validation, testing and obtaining the hyper-parameters of the model, we use the K -fold cross-validation algorithm (for more details see Camporeale, 2019). We use K values from 5 to 20 for the Bayesian inference model of sunspots to obtain predictions of sunspot Cycles between 25 to 32.

In this work, we have applied and modified the LS-SVM algorithms and toolbox by Suykens et al. (2005) to quantify the quality of the forecast estimates of the solar cycles, as well as to ensure that these cycles contain the information on the solar magnetic patterns found with the WT (see, Velasco Herrera et al., 2015, for more details).

3. Results and interpretations of wavelet analysis

In this section, we describe the time–frequency analysis of the Version 2.0 SSN record from 1700–2019 using the wavelet method.

3.1. 11-yr and longer solar cycles and patterns

Fig. 1 shows the characteristic patterns (periodicities) of the sunspot time series and their evolution between 1700 to 2019. The most prominent characteristic magnetic pattern of sunspots shown in the global wavelet (left panel, Fig. 1b) is the periodicity of the solar cycle at about 11 years known as the Schwabe sunspot cycle (Schwabe, 1844). This pattern is intimately related to different solar activity indices such as the solar flare index, the 10.7 cm solar radio flux, and the Total Solar Irradiance (TSI), among others. However, we caution that although SSN is a proxy that explains some of the variability of TSI, the SSN record should not be automatically presumed to represent TSI or other physical measures such as the Solar UV irradiance. Li et al. (2012) calculated that up to 43% of the daily variability in TSI can be explained by SSN over the 1979–2010 interval and Xu et al. (2017) reported the changing time lags between SSN and TSI over Cycles 21 to 24.

Evolution of the 11-yr solar cycle pattern is shown in the Morlet wavelet spectrum (central panel, Fig. 1c). The spectral power varies significantly with time and particularly during secular low power phases (i.e., during the Maunder, Dalton and Modern lows), the 11-yr wavelet intensity decreases but does not disappear completely. During the oscillation of the secular maxima, the 11-yr wavelet intensity also increases. It has been suggested that the amplitude of the solar cycle is modulated by a solar multi-decadal-to-centennial-long or secular magnetic periodicity between 50

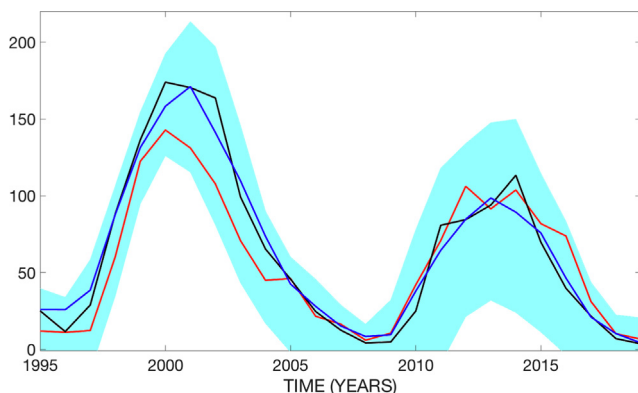


Fig. 4. A direct test of our hindcasting and forecasting methodologies. Comparison between the observed solar Cycles 23 and 24 from the SSN time series Version 2.0 (black line), the Bayesian model (blue line), and the NARX forecasting model (red line). The blue shaded area represents the 95% confidence intervals of the Bayesian model. We use the Bayesian model from 1700 to 1995 as inputs to NARX and LS-SVM algorithms for the “prediction” of Cycles 23 and 24 from the 1996–2019 interval. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

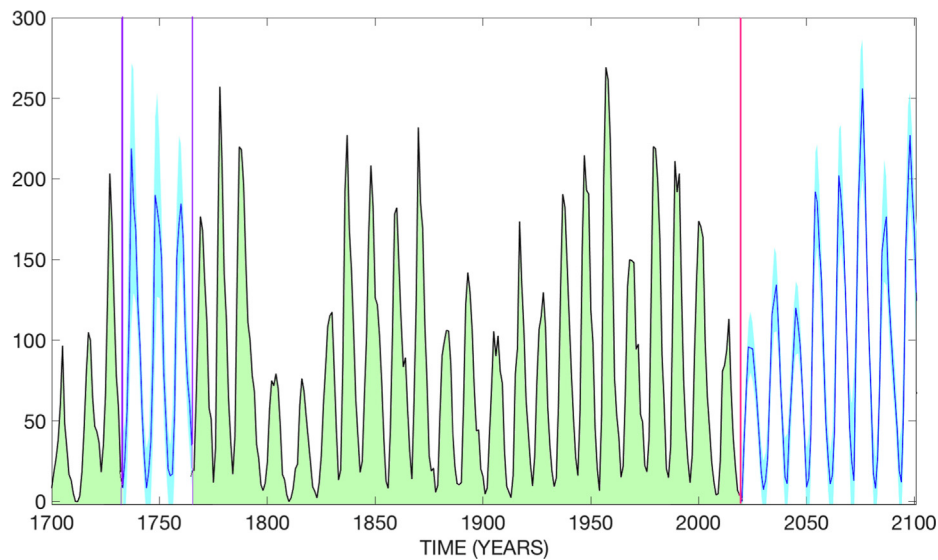


Fig. 5. Hindcasting and forecasting of sunspot cycles, 300 years backward (hindcast) and 100 years forward (forecast) based on the ML Bayesian inference of our empirically-derived LS-SVM model (separated by the vertical pink line a 2019). Our model hindcasts relatively higher sunspot numbers for Cycles -1 to 1 (bounded by the two purple lines) and extended low or weak future Cycles beginning from Cycle 24 and lasting until about 2050 (based on the power anomaly values shown in Fig. 6h). The blue shaded area represents the 95% confidence intervals of the Bayesian model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and 120 years (Gleissberg, 1939; Yoshimura, 1979; Lomb and Andersen, 1980; Feynman and Gabriel, 1990; Peristikh and Damon, 2003; Velasco et al., 2008).

On the other hand, Cameron and Schüssler (2019) casted doubts that these longer-term cycles are real. We propose that this issue has not been properly investigated thus far. Our wavelet analyses and empirical results show a longer-term (i.e., 120-yr) modulation cycle and we assumed that these cycles are real and applied it, in the sense of a working hypothesis, for the hindcasting and forecasting problems.

Ascertaining the solar magnetic secular periodicity that modulates the amplitude of the number of sunspots is necessary not only to forecast Cycle 25, but also for historical reconstruction. Amplitude of the solar sunspot cycle varies depending on its position within the secular oscillation phase. Based on this longer-term modulation, it is possible to forecast and hindcast beyond the limits of one solar cycle.

With different physical or mathematical models, distinct values can be obtained for the maximum of a sunspot cycle; however, not all forecasts are physically feasible or achievable. Regardless of which model is used, sunspot cycle prediction requires reproduction of the 11-yr solar cycle as well as solar magnetic patterns found in historical sunspot observations. It can be seen in Fig. 1d that the amplitude of the 11-yr pattern is modulated by the 120-yr longer-term oscillation (dotted red line, see below). We use this result not only to forecast the solar Cycle 25, but also to quantify how long the current grand maximum of solar activity will persist (Abreu et al., 2008) as well as when the new grand solar minimum will begin and how long will it last.

The second prominent magnetic pattern is a 120-yr periodicity with slightly less than a 95% confidence level (dashed red line in Fig. 1b) of detection in the formal sense of comparison with red noise spectrum (Gilman et al., 1963). This pattern is present throughout the entire time interval of the sunspot record and is related to the secular changes in the maxima and minima of solar activity (Velasco et al., 2008; Velasco Herrera et al., 2015). The 120-yr solar magnetic pattern is particularly apparent when analyzing the cosmogenic isotope records ^{14}C and ^{10}Be (i.e., as proxies of solar activity; see Velasco et al., 2008) and the signal also is detectable throughout the Holocene (see Soon et al., 2014). Based on the phase of this solar magnetic pattern, it is possible to quantify the beginning and ending of the secular solar maxima and minima.

Fig. 1g shows the 120-yr oscillation and the negative phase corresponds to the Maunder's minimum, Dalton's minimum, the Modern minimum and even the recent 21st century Minimum (respectfully, in time). The positive phases correspond to the secular maxima of solar activity around Maunder's maximum, Dalton's maximum and the Modern maximum phases, respectively.

One of the characteristics of this magnetic pattern is that the secular timescale at 120-yr globally modulates the amplitude of the 11-yr solar cycle (red dashed envelope in Fig. 1d). Furthermore, it is observed that during the 120-yr secular maxima (i.e., the positive phase in Fig. 1g), the amplitude of some sunspot cycles decreases (identified with the Roman numerals "I", "II", and "III" in Figs. 1a and 1d). This is surprising that a decrease exists in the amplitude of some solar cycles during secular maxima; no physical explanation can be posited. This pattern was originally observed by Rudolf Wolf (Wolf, 1862) and so we refer to

these phases “I–III” as Wolf’s cycles. Any long-term forecast of the sunspot series, in addition to reproducing the solar magnetic patterns, must also account for these Wolf’s cycles.

The secular magnetic pattern explains a portion of the complex variation/modulation in the amplitude of sunspots including the occurrence of Wolf’s cycles, since they occur on average every 120 years. Secular minima may simply correspond to the phases when the power of the solar dynamo falls below its average power, such as during Maunder’s minimum, Dalton’s minimum and Modern minimum. We note that the nature of this complex interaction will be better clarified in the nature of the 5.5-yr oscillation and its interaction with the 120-yr oscillation (presented below).

The third pattern observed in the global wavelet is the 60-yr periodicity (i.e., the Yoshimura–Gleissberg cycle; see [Soon et al., 2014, for an explanation and references](#)). The dynamics of 60-yr spectral evolution shows that it is very intense between 1700 to 1860. Then, the 60-yr-scale power decreases substantially until 1950 where it stabilizes between 1950 and 2019. After 1950, the spectral power is still nominally quite weak when compared to the 1700–1860 period ([Fig. 1f](#)).

We propose that the modulation of this 60-yr solar activity variation could be related to the inertial motions of the Sun around the solar system barycenter. A periodicity of magnetic origin that modulates the solar cycle should not be attenuated because this has strong implications on the various solar phenomena and on solar activity itself. The 60-yr periodicity does not globally modulate the amplitude of the 11-yr solar cycle ([Fig. 1d](#)). However, it may modify the solar magnetic field in such a way that it might affect the number of sunspots, particularly when it is in antiphase with the maximum of the secular cycle of 120 years during the occurrence of the Wolf’s maxima. That is, gravitational effects might modulate the electromagnetic solar phenomena in some way (see [Stefani et al., 2020a](#), for some proposed physical mechanisms). [Stefani et al. \(2020\)](#) recently examined the phase coherence and phase jump for 11-yr Schwabe cycles covering the early Holocene interval of 10,000–9,000 cal. BP and concluded that tidally-synchronized solar cycles is a viable explanation of the observations.

The fourth magnetic pattern is the 22-yr solar magnetic polarity cycle (i.e., the Hale cycle). In sunspots, which is an unsigned index, the spectral power of this 22-yr pattern is very attenuated (see [Fig. 1c](#)), particularly among 1700 and 1770. [Fig. 1g](#) shows the oscillation of the 22-yr magnetic cycles. This magnetic pattern is always present although it has been suggested that the magnetic polarity laws for sunspots may have been subdued during the Maunder Minimum of 1645 to 1715 (see [Sokoloff and Nesme-Ribes, 1994](#)).

3.2. The 5.5-yr solar cycle and patterns

The fifth magnetic sunspot pattern, probably the most important for forecasting purposes, is marginally observed

in the global wavelet spectrum ([Fig. 1b](#) and marked by white line across [Fig. 1c](#)). This 5.5-yr periodicity is detected with power just above the 95% confidence level (see the highlighted results of the 5.5-yr oscillation in [Fig. 2](#)). Its spectral evolution may yield important information on the individual characteristics of each solar cycle such as the shape/asymmetry and energy/power content of sunspot cycles, unlike the 11-yr Schwabe cycle that provides mainly global information (i.e., amplitude and duration) on each sunspot cycle.

Given the importance of the spectral information of the 5.5-yr solar cycle, and only for visualization purposes, we high-pass filter the 11-yr and longer cycles to highlight the spectral evolution of the 5.5-yr periodicity ([Fig. 2](#)). The Morlet spectrum with a periodicity of 5.5 years ([Fig. 2](#)) is identical in shape to that in [Fig. 1](#), but only under different units of normalization and re-scaling. In other words, the signals for the 5.5-yr oscillation in sunspot activity is detectably real from our wavelet analysis.

This 5.5-yr period has been shown to exist in historical aurora records by [Silverman \(1992\)](#), albeit with an intermittent or time-interval dependent nature. [Djurovic and Paquet \(1996\)](#) found that the 5.5-yr period was prominent and is common to all three geomagnetic field, Earth’s rotation, and solar activity variations.

[Polygiannakis et al. \(1996\)](#), using an RLC model, analyzed the shape and related morphological properties of the sunspot cycles such as the Waldmeier effect and the fact that the 11-yr oscillation introduces an infinite number of harmonics with periods $11/n$ yrs; with $n = 2, 3, \dots$ etc. Indeed, [Polygiannakis et al. \(2003\)](#), using the wavelet “skeleton” spectra of the SSN record, shows that the 5.5-year periodicity is a harmonic of the solar cycle.

[Mursula et al. \(1997\)](#), adopting an RLC model of the SSN, performed spectral analysis of the modeled sunspot time series and suggested that the 5.5-yr period in solar activity might be simply an artifact of spectral analysis. But it is important to note that even these authors have not precluded the interpretation of the 5.5-yr period as the harmonic of the main 11-yr sunspot cycles which is our tentative interpretation in this paper. Results and interpretation by [Makarov et al. \(2001\)](#) for the 5.5-yr cycle, show that this sub-harmonic may be interpreted as the time lag between the large-scale magnetic field of the Sun and sunspot cycles.

[Usoskin et al. \(2006\)](#), in analysing mainly the ^{10}Be cosmogenic isotope data as a proxy of solar activity, proposed that the 5.5-yr quasi-periodicity probably existed and can be interpreted as major Solar Proton Events that are more prevalent during solar activity maxima. This independent finding is clearly consistent with our results. [Velasco Herrera \(2008\)](#) reported on the relationship between the solar cycle and the secular solar cycle with the quasi-quinquennial periodicity of sunspots.

[Kollath and Olah \(2009\)](#) were able to confirm the existence of the 5.5-yr period in the SSN record by applying their methods of time series analysis that included the wavelet and short-term Fourier transforms. Likewise, these

authors also interpreted the 5.5-yr period as the harmonics of the dominant 11-yr sunspot cycles. Independently, [Le Mouél et al. \(2019a,b\)](#) also found a significant 5.5-yr signal in both the SSN and polar faculae activity records using the singular spectra analysis method. We agree that the nature and origin of the 5.5-yr period in the solar activity record needs to be more fully clarified but its existence does not appear to be in doubt.

In this paper, we revisit forecasts proffered by [Velasco Herrera \(2008\)](#):

“[T]he obtained results show that the 5.5 years (quasi-quinquennial) periodicity is systematically present in the spectrum of the secular maxima of solar activity. Such periodicity is attenuated or disappear[ed] during the secular minima: e.g. the Maunder, Dalton and Modern minima. Since such a quasi-quinquennial frequency has been attenuated during the preceding Cycles 22 and 23, therefore, we should expect that such a periodicity will be attenuated or disappear during Cycle 24 as well as in Cycles 25 and 26. Such a behavior will confirm that we are in the descending phase of the secular cycle toward its minima at the end of Cycle 26. Data of sunspots by the end of Cycle 24 will allow us either, to confirm our results on the relation between the 5.5 years periodicity and the solar secular cycle, or conversely to assume that the secular cycle is shifted, or even our results are not of general validity but only for some cycles as those associated to the Maunder, Dalton, and Modern minima.”

Wavelet transforms allow analysis at a specific periodicity and we can indeed perform such an examination for periodicities less than the 11-yr solar cycle ([Fig. 2b](#) and [c](#)). The global spectrum shows the periodicity of 5.5 years that exceeds the 95% confidence level (red dotted line in [Fig. 2b](#) and [c](#)) and its spectral evolution is shown in the Morlet wavelet spectrum ([Fig. 2c](#)). Amplitudes of the oscillation of 5.5 years ([Fig. 2d](#)) show that this solar periodicity is also modulated by the secular periodicity of 120 years (dashed pink curves). The spectral power of the 5.5-yr pattern is very intense and elevated during “high” solar cycles (with the exception of solar Cycles -1 to 1 , which are delimited by two pink vertical lines) and is significantly attenuated during “low” cycles and especially during the secular solar grand minima. This characteristic may be a key to predicting sunspot cycles, particularly Cycle 25. The 5.5-yr solar periodicity is not only an important manifestation of solar magnetic pattern, but could also be a precursor to the nature of solar activity cycle variations.

Before any forecast can be made, we must define a “high,” moderate, or “low” sunspot cycle. For this, we will use the statistical characteristics of the sunspot record ([Table 1](#)), which involves the mean value ($\overline{SSN}$) and its standard deviation (σ). These statistical parameters are shown ([Fig. 2a](#)) as a solid red line and two dashed red lines, respectively. One standard deviation above the mean value is labeled σ^+ , while one standard deviation below the mean value is labeled σ^- . Note that given SSNs are bounded from

below by zero (which can be attained in the low sunspot activity period), σ^+ and σ^- will be considered differently in the following discussion.

Under this nomenclature, a solar cycle is “high” if its amplitude is $\geq \sigma^+$, it is moderate if $\overline{SSN} < \text{amplitude} < \sigma^+$ and it is “low” if its amplitude $< \overline{SSN}$. One characteristic ([Fig. 2a](#)) is that the number of sunspots in the minima of each solar cycle is near or at about σ^- . So, σ^- can be considered as an upper limit on the minimum of each solar cycle.

We note that the statistical characteristics (i.e., the mean value and the standard deviation) will depend on the number of solar cycles included or the time interval of the observations. Use of these parameters to define “high” or “low” solar cycles is necessarily for convenience. Later, we will introduce a physical concept, such as temporal power anomalies (see [Velasco Herrera et al., 2015](#); [Soon et al., 2019](#)) that does not permit ambiguities but allows quantification of not only the intensity of solar cycles but also of the secular solar magnetic cycles.

[Hoyt et al. \(1994\)](#) inquired whether Rudolf Wolf’s sunspot reconstruction is correct and, after more than a quarter of a century, this question is still relevant and unanswered ([Hoyt and Schatten, 1998a](#); [Hoyt and Schatten, 1998b](#); [Clette et al., 2014](#)). An important task is to incorporate these data to define the accuracy of the sunspot reconstruction.

Until [Clette et al. \(2014\)](#), the original Wolf SSN time series was not significantly modified. Indeed, the first few sunspot cycles reconstructed by Rudolf Wolf are not only based on historical sunspot observations by keen observers, but were indirectly deduced or verified based on geomagnetic variations (i.e., magnetic needle readings). Thus, an important question is: How accurate are the sunspot cycles during the early to middle part of the 18th century in the Wolf reconstruction?

[Hoyt and Schatten \(1998a\)](#) cautiously revealed that although annual sunspot numbers are available from 1700, “Years with no value or an unreliable value [i.e., an uncertainty $> 25\%$] are ...1723, 1724, 1731, 1732, 1734, 1737, 1738, 1739, 1741, 1743, 1744, 1745, 1746, 1747, 1748, 1759, ...” (see [Fig. 1](#) in [Hoyt and Schatten, 1998b](#)). In reconciling past observations with modern telescopic observations, [Hoyt and Schatten \(1998a\)](#) highlighted the fact that “...the observations from 1653 to 1730 and from 1797 to the present are internally self-consistent to within 5%. Derived values between 1731 and 1796 are probably only self-consistent with modern observations to about the 15% to 20% level. Without the discovery of more observations, it will be difficult to reduce these errors” (see [Fig. 5](#) in [Hoyt and Schatten, 1998b](#)).

Even more directly relevant to the sunspot number record, [Clette et al. \(2014\)](#) noted that “after the Maunder minimum in the first half of the 18th century, the data coverage is particularly bad in the fifteen years from 1734 to 1748 [i.e., Cycles -1 and 0], during which the average

annual number of days with records is only 12. This lack of interest in systematic observations of sunspots is common during the 18th century. ...After the years 1744–1747 (without sunspot records), astronomers of the second half of the 18th and early 19th century also showed a limited interest for sunspot observations. Consequently, the yearly number of observations fluctuates around intermediate values, except for a new poorly observed interval from 1779 to 1794, before finally rising to almost 100% coverage in the early 1810s” (see Figs. 11 and 14 in Clette et al., 2014).

Indeed, we have no physical reason to explain the attenuation of the spectral power of the 5.5-yr periodicity in solar Cycles -1 to 1 (Fig. 2c), which should be classified as “high” activity. But considering Hoyt and Schatten (1998a,b) and Clette et al. (2014), we propose substantial missing sunspot counts for Cycles -1 to 1 (Fig. 2a,c,d and e). The lack of direct sunspot observations during this key interval (ca. 1733–1767) has also been cautioned early on by Letfus (2000), Vaquero (2007), and others (see Figs. 2 and 3c in those papers, respectively).

The primary efforts in reconstructing sunspot time series focuses on fluctuations and variability of the amplitude of the primary 11-yr solar cycles. In addition to the 11-yr periodicity, reproducing the solar cycle is a necessary but not a sufficient condition in that the reconstructed solar cycles must also comply with the 5.5-yr solar magnetic pattern (Fig. 2). With the lack of solar information noted for the first few sunspot cycles, it is necessary not only to forecast the sunspots of the following solar cycles, but also to accurately hindcast the first few solar cycles with ML. This effort will be discussed below.

The complexity of each solar cycle (such as amplitude, duration, asymmetries/shapes, among other parameters) makes it difficult to compare them with each other. To a first approximation, statistical characteristics merely suggest that a solar cycle is “high” or “low”. To overcome this simple categorization in comparing individual solar cycles, we define the temporal power index (see Velasco Herrera et al., 2015; Soon et al., 2019, for more details) of each solar cycle. This temporal power index is used as a proxy indicator of the energy released by the solar dynamo and hence the level of activity for each 11-yr solar cycle.

Fig. 2e shows the temporal power anomalies for each individual 11-yr solar cycle. If the anomaly is positive, then it is a very active solar cycle, while if the anomaly is negative, the solar cycle can be categorized as low or weak. Temporal power anomalies are negative during the secular minima (i.e., the Maunder, Dalton and Modern Minima which coincide with the negative phases of the 120-yr solar oscillation) and are positive around the secular maxima (i.e., the positive phases of the 120-yr solar oscillation). This information about the secular 120-yr high activity level is true but with the noted exception for solar Cycles -1 to 1 (i.e., the gray shaded area bounded by two vertical pink lines) explained above.

Anomalies of the temporal power of the solar cycles of the sunspots coincide with the spectral power of the

5.5-yr solar magnetic pattern; that is, positive anomalies of the solar cycle coincide with intense spectral powers (in red) of the 5.5-yr cycle (Fig. 2e). Negative anomalies of the solar cycle coincide with low spectral powers (in blue) of the 5.5-year cycle. Correspondence between these two solar magnetic patterns is surprising. In addition, the power anomaly of solar Cycle 23 is less than that of solar Cycle 22 and it has been noted that solar Cycle 23 is magnetically weaker than Cycle 22 (Bonev et al., 2004). This result may indicate the connection between the power anomalies and the magnetic activity of each solar cycle.

The lack of solar information on early sunspot cycles leads to negative anomalies of their power. These anomalies should be positive since they occur during the positive phase of the 120-yr solar magnetic pattern.

Judging from both the negative anomaly power of solar Cycle 24 and the negative phase of the 120-yr solar magnetic pattern for the 21st century, we can classify both Cycles 24 and 25 as moderate or low. From this basis, we can confidently conclude that solar Cycle 25 cannot be high (i.e., it cannot be much greater than σ^+).

According to the 120-yr solar magnetic pattern, a new great solar minimum should start at the beginning of the 21st century. Moreover, the strongest negative power is expected around 2040 as first proposed by Velasco et al. (2008). Next we can discuss our hindcast and forecast of past and future of solar activity, respectively, using the various ML algorithms to quantify and model sunspot activity cycles.

4. Results of hindcasting and forecasting solar activity cycles

4.1. Quality criterion of the Bayesian sunspot model

Because solar activity is a highly variable, complex system, and because ML model estimates are limited by an uncertainty principle (Velasco Herrera et al., 2015), only probabilistic ranges can be specified for the annual average number of sunspots.

To create a unique model for forecasting future solar cycles, the model must have (a) information on the patterns of the sunspot cycles found in the spectral analysis, as well as (b) information on variability that induces solar activity maxima and minima. A model with an artificial intelligent algorithm can never reproduce 100% of the objective data because this scenario leads to undesirable overtraining.

Here, we show that with the analyzed sunspot time series, a ML model can be constructed that meets both the requirements (a) and (b). We used the Bayesian model firstly to obtain a model that can accurately hindcasts sunspot patterns and can be used in forecasting sunspot counts over the next several cycles.

Since it is not possible to forecast the exact number of annual sunspots, we have applied the Bayesian ML model to reproduce the patterns of annual sunspots for forecasting subsequent solar cycles.

We use the mean annual sunspot number from 1700 to 2019 as inputs to a Bayesian inference model to yield a probabilistic model of sunspots activity variation. The correlation coefficient (ρ) and the mean squared error (mse) between the sunspot time series records and the ML Bayesian model are $\rho = 0.98$ ⁴ and $mse = 145.86$ sunspots, respectively. Furthermore, the standard deviation (σ_m) and mean value (μ_m) of the ML Bayesian model are $\sigma_m = 54.9$ and $\mu_m = 78.6$ sunspots. These results show that the ML Bayesian model does not suffer from over- or underfitting that usually affect machine training methods. Therefore, it is reasonable to apply this ML model to professional scientific forecasts of solar activity.

To evaluate the performance of our Bayesian model of sunspots from 1700–2019, we employ the methods outlined by Legates and McCabe (1999, 2013) who recommended using both absolute and relative error methods. Absolute error measures are given in the units of the variable; here, in annual number of sunspots. Two absolute error measures are often reported mean absolute error (MAE; the average difference between the predictions and the observations) and the root mean square error (RMSE; the square root of the sum of the squared differences between the predictions and the observations). RMSE is more responsive to the outliers than MAE but both are widely used (Legates and McCabe, 1999).

For the Bayesian model, the MAE is 8.798 while the RMSE is 12.078. This indicates that the model is able to predict the observed sunspot count to within an average of less than nine sunspots while the larger value of RMSE suggests the model tends to exhibit more uncertainty at the extreme values; generally, the peak of the sunspot cycle.

Legates and McCabe (1999, 2013) further suggest a relative error measure should be reported with model evaluation. They recommend the Nash–Sutcliffe statistic, NS_k , given by

$$NS_k = 1 - \frac{\sum_{i=1}^N |O_i - P_i|^k}{\sum_{i=1}^N |O_i - \bar{O}|^k} \quad (6)$$

where k is the exponent of the Nash–Sutcliffe statistic, O_i and P_i are the observed and Bayesian-predicted yearly sunspot counts, respectively, and N is the number of observations (here, $N = 320$). The original Nash–Sutcliffe statistic uses $k = 2$ although Legates and McCabe (1999) argue that $k = 1$ is a better choice as it directly parallels MAE. NS_k varies from 1.0 (i.e., a perfect model) to $-\infty$; the value of NS_k decreases as the model performance degrades. A value of $NS_k = 0.0$ implies that the model is no better at predicting the observations than the observed mean (i.e., the *a priori* model assuming no ability to predict whatsoever).

Positive values, therefore, suggest some predictive ability while negative values imply the model is worse than simply using the mean sunspot number.

For the Bayesian model, $NS_2 = 0.9620$ while $NS_1 = 0.8275$. Generally, $NS_1 < NS_2$ but both statistics suggest that the model adequately resolves the variability in annual sunspot numbers to a very high degree. This result is encouraging.

However, Legates and McCabe (1999, 2013) have criticized the use of Eq. (1) without considering the *a priori* model. As originally formulated, the Nash–Sutcliffe statistic compares the model against the predictive ability of the observed mean. If a strong periodicity or seasonality exists in the data, then the model should not be “rewarded” for being able to resolve that trend. Here, the 11-year cycle is a strong signal that any model should easily be able to resolve.

Instead of comparing against the observed mean, therefore, $\bar{O}$ in Eq. (1) should be replaced by a model that simply predicts a cosine wave with a periodicity of 11 years (i.e., the dominant Schwabe cycle), a phase shift commensurate with the peak of the sunspot cycle, and a constant amplitude defined by the average amplitude of the observed record. This “simple model” (SM) is given by:

$$SM(year) = 47.72 \cos(0.5712 * year - 0.3662) + 78.735 \quad (7)$$

When compared to this simple 11-year cosine function model, the Bayesian Model performs admirably. Its statistics are $NS_2 = 0.9459$ while $NS_1 = 0.7773$. This indicates that the Bayesian Model is very well able to simulate the differences in the magnitude and timing of the sunspot cycles and has much more efficacy than a model that simply reproduces only the 11-yr cycle.

Fig. 3 shows that indeed the ML Bayesian model (blue line) reproduces the patterns of the annual sunspot time series records from 1700 to 2019 with the 95% confidence intervals also indicated (blue shaded area). In particular, we highlight that due to these confidence levels, the Bayesian model shows that solar Cycles 0 and 1, were actually higher than σ^+ and more similar to solar Cycle 2, while solar Cycle -1 appears to be similar to solar Cycle -2 . In other words, these three solar cycles (Cycles -1 to 1) should be comparable to solar cycles that are in the positive phase of the 120-yr solar cycle – their sunspot numbers should be greater than σ^+ .

4.2. Forecasting and hindcasting of solar cycles

Before any forecast of the next several solar cycles can be made, we quantify the ability of our NARX model to “predict” recent solar cycles. As an example, we used as input data of the Bayesian model obtained from the Version 2.0 SSN time series between 1700 and 1995 to test the reproduction of the observed solar Cycles 23 and 24. Fig. 4 shows the comparison between solar Cycles 23 and 24 from the Version

⁴ Note, however, the correlation is extremely high simply because much variation exists between the peak and the minimum of the sunspot cycle and the model is able to resolve the dominant 11-year Schwabe cycle

2.0 SSN time series (black line), the Bayesian model (blue line), and the NARX forecasting-model (red line). We note that applying the algorithm described in Section 2.3.3, the NARX forecasting-model (red line) reproduces the evolution of the observed solar Cycles 23 and 24 (black line) reasonably well and that the forecasted solar Cycles 23 and 24 are within the blue shaded area that represents 95% confidence intervals of the Bayesian model. This result has an MSE of 120.3 and a linear correlation coefficient of 0.9 and it can be confirmed that training the NARX using data from the Bayesian model reproduces the key characteristics of solar Cycles 23 and 24.

Given the ability of NARX to hindcast the recent two solar cycles, we will now forecast consecutively the next two solar cycles starting at Cycles 25 and 26 and continuing through solar Cycles 31 and 32 using the algorithms and steps described in Section 2.3.3. The goal in forecasting these solar cycles (i.e., from 25 to 32) is to see whether the next Grand Minima might exist; it has been speculated to begin sometime during the early 21st century. In addition, a second NARX model developed with the information from solar Cycles 24 to 2, progressing backward in time, is used to carry out a scientific hindcast of the past solar Cycles from -1 to 1 . We undertook this hindcasting experiment based on our Fig. 2 as well as the possibility of missing or uncertain sunspot reconstructions for these three early solar Cycles as reported by Hoyt and Schatten (1998a,b) and Clette et al. (2014).

Five hundred probabilistic forecasts were used to provide confidence levels for forecasts of solar Cycles 25 to 32. A second Bayesian inference of LS-SVM then was used to get a ML model of solar Cycles 25 to 32 (to the right of the vertical pink line in Fig. 5) to within a 95% level of confidence (blue shaded areas).

We repeated this algorithm with the inverted time integration of the solar cycles (i.e., starting from Cycle 24 and proceeding back to solar Cycle 2) to hindcast solar Cycles from -1 to 1 (between the two purple lines in Fig. 5). Fig. 5 also shows the historical sunspot time series from 1700 to 2019 (green area) and the ML forecasts of sunspot time series for solar Cycles 25 to 32 (blue shaded areas) as well as the hindcast sunspot time series for solar Cycles -1 to 1 (i.e., blue shaded areas between two purple lines).

We can also observe (Fig. 6) that the ML model reproduces the pattern of the first 24 solar Cycles in the wavelet domains. In addition, the number of forecasted sunspots during the minima of these Cycles (i.e., from 25 to 32) is less than σ^- , similar to what occurred during the past solar cycles for those known secular minima or the Grand minima. Our quantification of such forecasted minima is presented (Fig. 6) through the wavelet spectral analysis (as in Figs. 1 and 2).

5. Discussion

The SSN time series can be classified as nonstationary. Its stochastic/chaotic component makes sunspot numbers

unpredictable on relatively long timescales. Accurate predictions for several cycles ahead are not possible owing to the random fluctuations with nonlinear effects (see e.g. Kremliovsky, 1994; Kremliovsky, 1995; Charbonneau, 2020; Petrovay, 2020).

Different studies provide arguments against long-term predictability (see e.g. Solanki and Krivova, 2011) owing to the non-linearity of the solar dynamo. Charbonneau (2020) argues that a mathematical model used to study a physical system that cannot be understood by the observed data should not be elevated to “physical realism” since a model is not reality. If the dynamo model cannot explain the complex behavior of sunspots, then it cannot predict solar activity. By contrast, Bouwer (1992) and Mendoza and Velasco Herrera (2011) suggest that solar activity probably originated from chaotic quasi-periodic processes and not from stochastic or intermittent processes.

Steinhilber and Beer (2013), in forecasting solar activity for the next 500 years, noted that arguments against long-term forecasts have not accounted for information based on cosmogenic isotopes records that show stable periodicities throughout the Holocene (see Soon et al., 2014). Based on this multi-millennia solar activity behavior, solar magnetic patterns (as deduced empirically from both the sunspot record and proxy solar activity records) will continue to drive sunspot activity (Steinhilber and Beer, 2013). It is the 120-yr periodicity that has been reported throughout the Holocene (Soon et al., 2014) which has figured prominently in our study.

The greatest challenge in solar forecasting is predicting the amplitude and duration of the next solar cycle (Hathaway, 2015; Petrovay, 2020) and determining its level of accuracy. If we assume that each solar cycle has an intrinsic energy and a given frequency (i.e., with sunspot activity periods between 8 to 14 years), the accuracy of the estimated models of the solar activity should be limited by an uncertainty principle (Velasco Herrera et al., 2015). Therefore, it is indeed impossible to predict any parameter of the solar cycle of sunspots (e.g., phase, amplitude, and period). As hinted in the discussion above, we strive to establish a probabilistic interval in which the sunspot estimates may fluctuate with a high level of confidence and verisimilitude.

Gleissberg (1942, 1945) remark that the fluctuations of the sunspot cycles are not regular enough to obtain an analytical function. Not only are the behavior of future solar cycles impossible to predict, but exact predictions of the number of sunspots are impossible. Thus, the problem of sunspot cycle predictions should be considered as a question of probability. This is why the solution to solar cycle forecasting consists in calculating the prediction probabilities of future solar cycles. Camporeale (2019) suggests that the great challenge in our contemporary study of space weather is to change the forecasting paradigms from point forecasts to a probabilistic approach with reliable uncertainties.

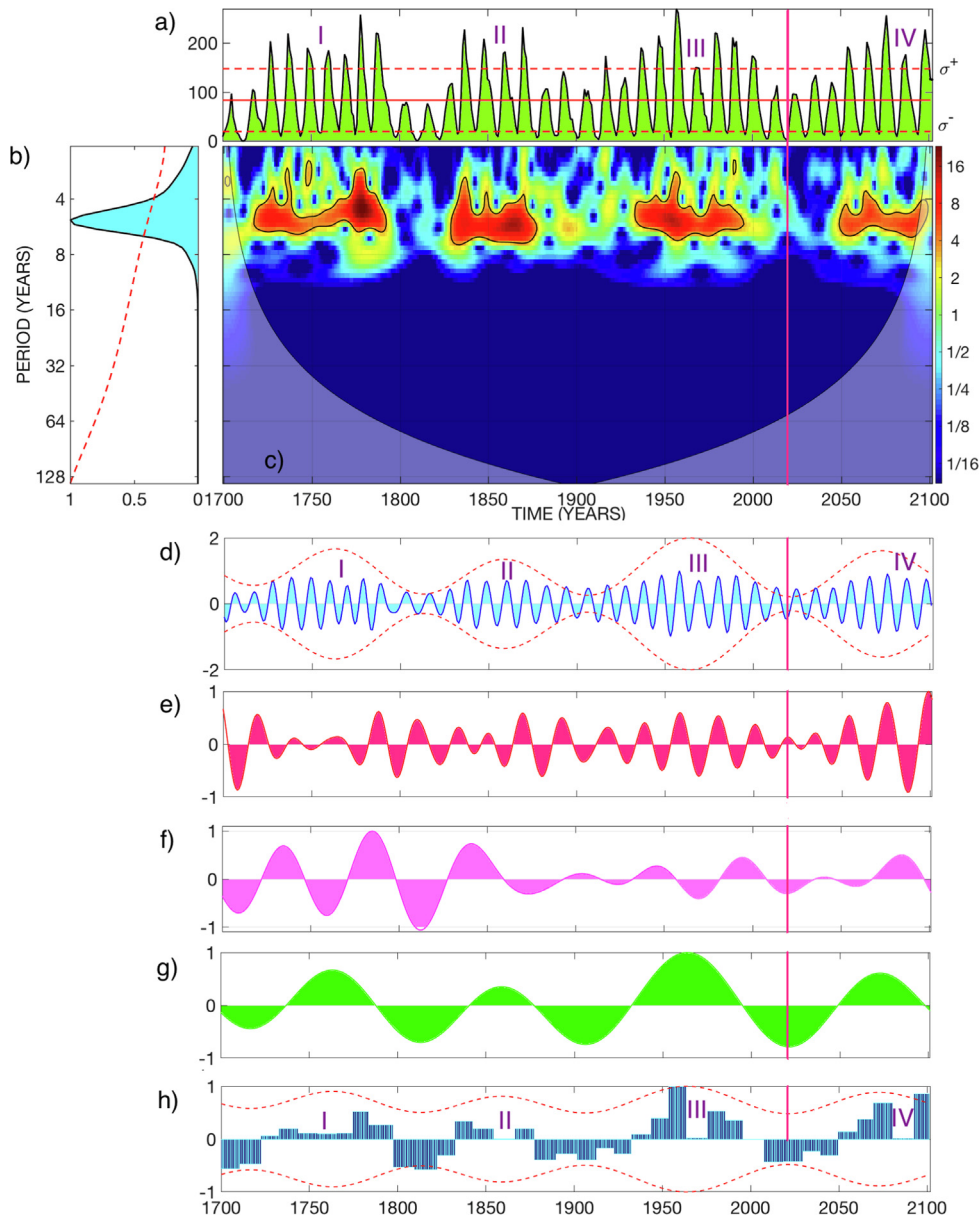


Fig. 6. Wavelet analysis of the ML-derived sunspot time series from 1700–2100 shown in Fig. 5. Panels (a), (b) and (c) represent information as described in Fig. 2. Panels (d), (e), (f), (g), and (h) are the filtered time series for the 5.5-yr, 22-yr, 60-yr, 120-yr periods and the 11-yr cycle power anomaly, respectively. Dashed pink lines in (d) and (h) are the envelope of modulation by the 120-yr oscillations. The vertical pink line denotes the year 2019 (i.e., present-day). Roman numerals “I”, “II”, “III” and “IV” in panels (a), (d) and (h) marked. Wolf’s maxima. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Our Bayesian model forecasts a peak annual sunspot number of 95 ± 15 for Cycle 25. But as our model is probabilistic, the maximum sunspots vary between 80 to 115 sunspots with the maximum expected around 2023 to 2025. Others have published forecasts for solar Cycle 25 and peak sunspot numbers vary from very low to extremely high values. This underscores the uncertainty associated with solar cycle prediction.

Kitiashvili (2020), adopting synoptic magnetograms for forecasting of solar activity, suggested that the Solar Cycle 25 will be weaker than the solar Cycle 24. The sunspot maximum will occur in either 2024 or 2025 with a maxi-

mum sunspot number of $\sim 50 \pm 15$. This SSN would be even less than the number of sunspots during the Dalton minimum and possibly comparable to the Maunder minimum. However, no physical indications exist that the next great solar minimum is as low as the Maunder minimum. We have carried out a new reconstruction of the SSN for the entire Holocene to analyze the Grand solar minima similar to the Maunder-type and the results will be discussed in another paper.

Bhowmik and Nandy (2018), using a solar surface flux transport model and an internal solar dynamo model, provides a prediction of yearly mean sunspot number of 118 at

the maximum of solar Cycle 25 with a range from 109 to 139, and peak date of $\sim 2024 \pm 1$. Also, they suggest the possibility of a somewhat stronger solar Cycle 25 than expected; that is, it does not indicate a substantially weaker solar Cycle 25 compared to Cycle 24 and perhaps even reverses the substantial weakening trend in solar activity. These authors suggested that this rules out a weak Maunder-like minimum in solar activity. According to the solar power deficit of sunspot Cycle 24 and the depth of negative power anomalies of sunspot Cycles 25–27 (Fig. 6h) obtained with ML algorithms, we propose that the weakening in solar activity will not end immediately after solar Cycles 24 and 25. Our Bayesian model estimated that the secular sunspot minimum of the 21st century could be classified to be similar in intensity to the Dalton Minimum.

McIntosh et al. (2020) have adopted physical modeling of the cycles in magnetic activity and have predicted Cycle 25 to be “among the strongest sunspot cycles ever observed” with a peak SSN amplitude of 233 sunspots and one standard deviation confidence interval between 204 and 254 sunspots). Finding and confirming the solar magnetic secular periodicity that modulates the solar cycles of SSN is necessary to forecast Cycle 25. The amplitude of the solar sunspot cycle will vary depending on its position within the secular oscillation phase. Based on this modulation, it is possible to qualitatively forecast the amplitude of the solar Cycle 25. Different physical or mathematical models yield different values for the maximum of solar Cycle 25. However, not all forecasts are physically feasible and achievable. In accordance with both the negative power anomaly of solar Cycle 24 and the negative phase of the 120-yr solar magnetic pattern for the 21st century, we classify solar Cycle 24 as moderate or low. From this basis, we suggest that solar Cycle 25 cannot be high (i.e., it cannot be much greater than $\sigma^+ = 141$ sunspots, Fig. 2a and e). Our ML model, in addition to predicting a great Dalton-type solar minima for Cycles 24–27, also predicts another great solar maximum from solar Cycle 28 onward. It is predicted that solar Cycle 30 would be comparable with the well-known high SSN peak for solar Cycle 19; i.e., a solar cycle with more than 200 sunspots (e.g., see Fig. 6a).

Results of the theoretical forecast (Table 2) of the last two solar minima (Dalton and Modern) from the Maunder minimum (1645–1715) apply periodicities of 120, 88 and 100 years. To obtain the theoretical Dalton minimum for these three modulating periodicities, the start and end times of the Maunder minimum (1645–1715) interval are added forward in time by 120, 88 and 100 years and we obtain 1765–1835, 1733–1803 and 1745–1815, respectively, for estimated dates of the Dalton Minimum. Far more uncertainties exist for the 88 and 100-yr modulation periodicities. Theoretical hindcasting using the modulation periodicity of 120 years yields the actual reported interval of the Dalton Minimum (1790–1830).

Table 2

Hindcast/forecast of historical solar minima and the new solar minimum from the Maunder minimum (1645–1715) onward under different assumed solar magnetic patterns that could modulate the amplitude of sunspots. The start (1645) and end (1715) year of the Maunder Solar Minimum is added forward with the value of 120, 88 and 100 years and the Dalton minimum are then “predicted”. The interval of the theoretical Dalton minimum is then again increased by 120, 88 and 100 years to obtain the theoretical interval of the Modern solar minimum. Finally, the theoretical interval of the Modern minimum is again increased by 120, 88 and 100 years to obtain a theoretical forecast of the new solar minimum in the 21st century. Only the 120-year period correctly predicts the theoretical start for the new solar minimum in the 21st century which agrees with many of the cited publications. By contrast, the secular modulation periods of 88 and 100 years predict the end of new solar cycle by the 21st century.

Solar Minimum	(Start–End)	120-yrs (Start–End)	88-yrs (Start–End)	100-yrs (Start–End)
Maunder	(1645–1715)	(1645–1715)	(1645–1715)	(1645–1715)
Dalton	(1790–1830)	(1765–1835)	(1733–1803)	(1745–1815)
Modern	(1890–1939)	(1885–1955)	(1821–1891)	(1845–1915)
XXI		(2005–2075)	(1909–1979)	(1945–2015)

Applying this same methodology to the theoretical interval obtained from the Dalton minimum for each of the periodicities of 120, 88 and 100 years, we obtain intervals for the Modern minimum of 1885 to 1955, 1821 to 1891, and 1845 to 1915, respectively. Similarly, theoretical forecasts for the new solar minimum become 2005 to 2075, 1909 to 1979, and 1945 to 2015, respectively.

We note that the theoretical predictions using the 88 and 100 yr modulation periodicities have incompatible errors with the actual time intervals reported for the Dalton and Modern solar minima. Regarding the forecast of the new solar minimum, these periodicities suggest it has already ended. This exercise shows that secular periodicities of 88 and 100 years cannot accurately modulate sunspot activity. We consider that the Gleissberg periodicity is not 88 years (seven or eight sunspot cycles; Gleissberg, 1939) but is 120 years. The 120-yr periodicity forecasts yielded a strong agreement with the reported solar minima (Hathaway, 2015; Petrovay, 2020) and predicted the start of the new solar minima will extend from 2005 until 2075. This simple arithmetic deduction is compared with our more sophisticated deduction using the ML algorithm and the actual realization of the 21st century minimum of approximately 2008 to 2050 using the actual power anomaly metric (Fig. 6h). Our model correctly deduced the start of the 21st century sunspot minimum but cannot accurately suggest its ending. We attribute the discrepancy of ~ 25 years to the fact that solar activity originates from chaotic quasi-periodic processes and because it is impossible to predict the amplitude and period of a solar cycle according to the uncertainty principle.

We have shown the ability of ML using NARX algorithms to forecast sunspot activity from Cycles 25 to 32 and the ability to reproduce solar magnetic patterns from 1700 to 2019. Also, the sunspot forecast from ML is within

the probabilistic range of sunspot fluctuations with a high level of confidence and plausibility. We thus conclude that the periodicity of 11 years yields global information on sunspot cycle (amplitude and duration) while the subharmonic of 5.5 years provides information on each solar cycle (asymmetry/shape and energy/power content). The 120-year cycle appears to be the robust modulator of sunspot activity cycles beyond the 11-year timescale.

To avoid any confusion, we declare that we have adopted the interpretation of this 5.5-yr periodicity to be the main harmonic of the 11-yr sunspot cycles. This interpretation is in full agreement with the in-depth study by Polygiannakis et al. (2003) and Kollath and Olah (2009). Furthermore, we must note that our working hypothesis concerning this periodicity lies with the actual hindcasting and forecasting outputs when compared with actual reality but not regarding whether the periodicity is “fundamental” or not as argued in Mursula et al. (1997).

6. Conclusions

Characteristics of the next solar cycle are important parameters to estimate. Their knowledge is important for preparedness in space exploration, and in minimizing risks in technological activities. The ability to understand historical solar magnetic parameters more accurately, beyond the highly dominant ~ 11 -year cycle, is a key to estimating solar variability.

Not all estimates for the SSN maxima are physically feasible nor realizable. In predicting sunspot cycles, regardless of which model is used, it is not only important to reproduce the 11-yr solar cycle, but also all solar magnetic patterns found in historical observations. We have shown that the 5.5-yr periodicity is a fundamental solar magnetic pattern in forecasting one or more solar cycles ahead since this solar periodicity is coupled with other fundamental periodicities of solar activity (11, 22, 60 and 120 years). Furthermore, depending on its temporal position within the phase of the 120-yr oscillation (positive or negative), the anomaly of its spectral power will be high or attenuated, respectively.

We have presented a ML model of the variation of sunspot activity based on magnetic solar patterns and a methodology relying on the 5.5-yr and 120-yr periodicities. By analogy to the Waldmeier effect (i.e., the most active sunspot cycles are the most asymmetric), most active solar cycles are detected with very strong spectral power at the 5.5-yr periodicity and vice versa for the weaker solar cycles. One of the main aspects of the Waldmeier effect is that the discussion is centered on the shape of the solar cycle. But the shape form of a sunspot cycle is a mere subjective aspect. To overcome this deficiency, we have adopted the metric of power content (Feynman et al., 2011) in our study. Therefore, the subjectivity of the cycle shapes can be eliminated by using the temporal power anomaly of each solar cycle shown under Figs. 2e and 6h.

Importantly, the periodicity of 11 years quantifies global information on the sunspot cycles (amplitude and duration) while its sub-harmonic of 5.5 years provides detailed information on each solar cycle (asymmetry/shape and energy/power content of sunspot cycles).

Our model yields a reliable reconstruction of the sunspot record observed from 1700 to 2019, as well as probabilistic forecasts of the sunspot time series from 2020 to 2100. A promising result of this unique vision is our proposal to reevaluate and reconfirm sunspot numbers for solar Cycles -1 to 1 at around ca. 1733 to 1767. Our ML model forecasts that a new phase of extended solar minima that have started since Sunspot Cycle 24 (ca. 2008 to 2019) and will persist until Sunspot Cycle 27 (2050 or so).

Based on the solar power deficit of Sunspot Cycle 24 and on the depth of negative power anomalies of Sunspot Cycles 25–27 (Fig. 6h) obtained with ML, it is estimated that the secular sunspot minimum of the 21st century could be similar to the Dalton Minimum.

Our model forecasts a peak annual sunspot number of 95 for Cycle 25 around 2024 ± 1 with an uncertainty between 80 to 115 sunspots. In addition, we found that our ML pattern recognition method yields insights regarding the missing sunspots during the early sunspot records around Cycles -1 , 0 and 1 (ca. 1730s–1760s). This specific reality of under-estimating sunspot counts around this historical interval has long been documented by the pioneering sunspot reconstruction effort by Hoyt and Schatten (Hoyt and Schatten, 1998a; Hoyt and Schatten, 1998b) as well as the latest research of Clette et al. (2014).

7. Data Availability Statement

We used the International Sunspot Number (SSN) Version 2.0 from the World Data Center Sunspot Index and Long-term Solar Observations (WDC-SILSO), Royal Observatory of Belgium, Brussels (<http://www.sidc.be/silso/datafiles>).

The Bayesian Model of Machine Learning of Missing Sunspots and the Forecasts Solar Cycles 25–31 is available in:

1. <https://doi.org/10.7910/DVN/U6B15L>
2. <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/U6B15L>

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank Dr. Ronan Connolly for linguistic help and Dr. Rodolfo Cionco for the first reading of our draft paper.

We further acknowledge the critical reviews of both referees which have led to improvements of our manuscript. V.M. Velasco Herrera thank Dr. Nelya Babynets for her valuable comments and suggestions and acknowledges the support from CONACyT-180148 and the support from PAPIIT-IT102420 grants. W. Soon's work was partially supported by the SAO grants with proposals ID: 000000000003010-V101 and 000000000004254-V101. D.R. Legates did not received any funding for this research.

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